TOWARDS A NEW WRITING STYLE

STYLOMETRIC DRIFT AND AUTHORSHIP IN HUMAN–AI CO-WRITTEN TEXT

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RESEARCH QUESTIONS

Main research question:

Can we detect and characterise the linguistic style of human-AI co-written text, and how does it differ from purely human or purely AI writing?

Sub-research question:

RQ1. *Can Human, AI-generated, and Human–AI hybrid texts be classified as distinct stylistic categories?*

RQ2. *How does linguistic style change as the level of AI rewriting increases?*

RQ3. *Does authorship attribution remain robust under AI editing?*

KEY CONCEPTS

- **stylometric drift** = in this study, the change in writing style between an original text and its AI-modified version (Tripto et al., 2024)
- **rewrite intensity** = in this study, the degree of AI intervention in a text (Li et al., 2024)

RELEVANCE

ACADEMIC INTEGRITY

AI tools challenge traditional definitions of authorship in education and publishing (Dwivedi, 2023).

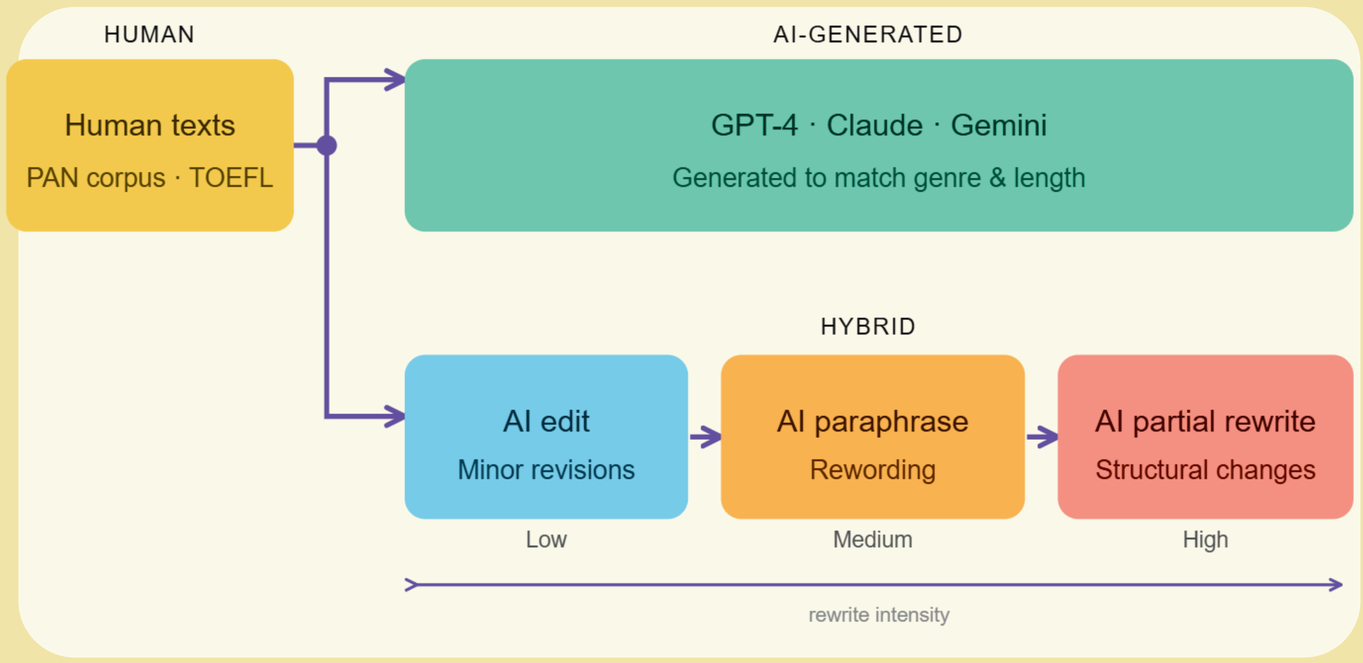
AUTHORSHIP VERIFICATION

AI rewriting may weaken stylistic signals used in authorship attribution (Tripto et al., 2024).

SCIENTIFIC CONTRIBUTION

Hybrid texts are expected to form a distinct stylistic category.

DATASET - THREE PART CORPUS



DATA SNIPPET

HUMAN

"Speculator greed and corruption was merely a subdivision behind the United States' financial dilemma; the common man was just as guilty."

Source: PAN'25/26 Generative AI Detection: Voight-Kampff AI Detection Sensitivity, train split, human essay

AI-GENERATED

"In the United States, economic crises often stem from reckless lending, weak oversight, and government failures that let financial problems build up too far."

Prompt: Write one sentence about the causes of an economic crisis in the United States.

Requirements:
· Length: 20 words · Tone: formal and slightly argumentative · Style: not perfectly polished · Sentence structure: moderately complex · Vocabulary: academic, not overly sophisticated · Causal explanation and mild judgment (assigning responsibility)

Resemble a student essay in history or economics.

HYBRID

"Speculator greed and corruption were only one part of the United States' financial crisis; ordinary people were also partly to blame."

Prompt: Paraphrase this sentence at a medium level of rewriting. Keep the meaning and tone, but reword awkward phrasing without fully rewriting it.

Sentence:
"Speculator greed and corruption was merely a subdivision behind the United States' financial dilemma; the common man was just as guilty."

STYLOMETRIC ANALYSIS

Surface-level features such as spacing, capitalization, and formatting are normalized during preprocessing, while generation is controlled for factors such as topic, genre, and approximate length. Analysis therefore focuses on deeper stylistic features such as:

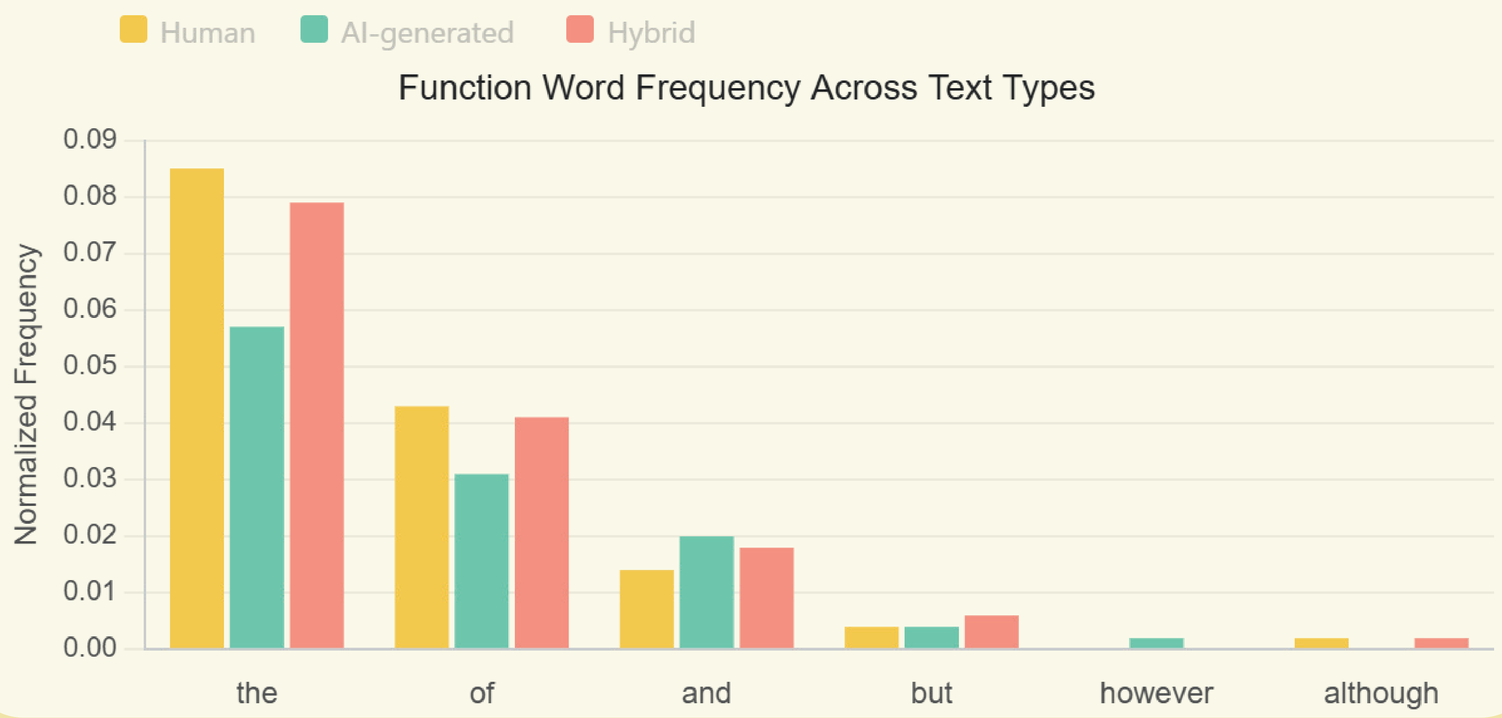
Lexical diversity (Type–Token Ratio)

Lexical choice (word selection: simpler vs more complex)

Function word frequency (the, of, and, but, however, although)

ILLUSTRATIVE RESULTS

Most features show a gradual stylistic shift as AI involvement increases, with hybrid texts often displaying intermediate patterns. If this trend generalizes, it would support the idea of stylometric drift.



References

Dwivedi, Y. K. (2023). "So What If ChatGPT Wrote it?" Multidisciplinary Perspectives on opportunities, Challenges and Implications of Generative Conversational AI for research, Practice and Policy. International Journal of Information Management, 71(0268-4012), 102642. https://doi.org/10.1016/j.ijinfomgt.2023.102642

Li, Y., Wang, Z., Cui, L., Bi, W., Shi, S., & Zhang, Y. (2024). Spotting AI's Touch: Identifying LLM-Paraphrased Spans in Text (pp. 7088–7107). https://aclanthology.org/2024.findings-acl.423.pdf

Tripto, N., Venkatraman, S., Macko, D., Moro, R., Srba, I., Uchendu, A., Le, T., & Lee, D. (2024). A Ship of Theseus: Curious Cases of Paraphrasing in LLM-Generated Texts. 1, 6608–6625. https://aclanthology.org/2024.acl-long.357.pdf